UNITED INTERNATIONAL UNIVERSITY

Department of Computer Science and Engineering

GRAMBANDHAN: SMART RURAL INVESTMENT PLATFORM

Project Management Report – Midterm Submission

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Course:	PMG 4101 Project Management
Date:	September 6, 2026

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1. Project Overview

1.1 Problem Statement

In rural sectors, farmers, livestock owners, and artisans often struggle to lead stable livelihoods due to a severe lack of financial support and resources. Traditional agricultural finance heavily relies on personal networks or local moneylenders, limiting access to adequate capital. Current systems force rural producers to sell through intermediaries who take large commissions, heavily reducing the earnings of the actual creators. Furthermore, urban investors interested in funding agricultural projects face a lack of transparency and an inability to track how their money is utilized, leading to widespread trust issues.

1.2 Vision

The vision of Grambandhan is to create a digital platform that effectively connects rural producers with investors, supporting farmers and artisans by providing financial opportunities and fair market access. The platform aims to reduce dependency on middlemen and establish a transparent, profit-sharing investment model that ensures sustainable rural development and mutual benefit.

1.3 Scope and Features

The Grambandhan system is scoped as a standalone web and mobile application designed to bridge the gap between rural agriculture and urban capital. The core features include:

- **Project Listing System:** Farmers can create and manage project listings with required funding, timeline, and expected outcomes.
- **Profit-Sharing Investment:** Investors can fund projects and earn based on outcomes instead of fixed interest rates.
- **Real-Time Progress Tracking:** Live updates with photos and reports enable investors to stay connected with the projects.
- **Online Marketplace:** A direct selling platform allowing farmers to sell agricultural products to buyers without intermediaries.
- **AI-Based Risk Suggestions:** Smart recommendations and predictions to reduce risks and improve outcomes.
- **Insurance & Risk Coverage:** Protection against unexpected losses through an integrated central insurance pool.
- **Fraud Protection System:** Enhances security through verified users, secure transactions, and a rating mechanism.
- **Women Empowerment Support:** Special resources and visibility aimed at supporting women in handicrafts and small businesses.
- **Digital Payment Integration:** Secure financial transactions utilizing mobile banking services like bKash and Nagad.

2. Methodology

2.1 What is Methodology

A project methodology is a systematically structured framework of rules, practices, and procedures used by a team to plan, execute, and deliver a project. It defines the workflow and lifecycle phases required to transform initial requirements into a final, deployable product.

2.2 Waterfall Model

2.2.1 Waterfall Model Design

A linear, sequential approach where each phase (Requirements, Design, Implementation, Verification, Maintenance) must be completed before the next begins.

2.2.2 Advantages of the Waterfall Model

Simple to understand and manage; clear milestones and rigid structure.

2.2.3 Disadvantages of the Waterfall Model

Highly inflexible to changes; late testing phase makes fixing core design issues expensive.

2.3 Scrum Development Methodology

2.3.1 Different Roles in Agile Scrum Methodology

Product Owner (represents stakeholders), Scrum Master (facilitates the process), and Developers (build the product).

2.3.2 Scrum Development Process – Design

Iterative development delivered in short timeboxes called “Sprints”, usually lasting 2-4 weeks, featuring daily stand-ups and sprint reviews.

2.3.3 Advantages of Scrum Development

High adaptability to changing requirements; continuous user feedback and early delivery of value.

2.3.4 Disadvantages of Scrum Development

Can lead to scope creep; requires highly experienced and self-organized team members.

2.4 Prototype Methodology

2.4.1 Prototype Model – Design

Involves building a working replica or early version of the system to demonstrate functionality, gather feedback, and refine requirements before final development.

2.4.2 Advantages of Prototype Model

Validates requirements early; highly effective in reducing design risks through immediate user interaction.

2.4.3 Disadvantages of Prototype Model

Can be costly and time-consuming; users may confuse the prototype with the finished, fully-functional product.

2.5 Incremental Model

2.5.1 Incremental Model – Design

The product is designed, implemented, and tested incrementally, adding functionality module by module until complete.

2.5.2 Advantages of Incremental Model

Core functions are delivered early; easier to test and debug smaller iterations.

2.5.3 Disadvantages of Incremental Model

Requires excellent upfront planning and a well-defined architecture to ensure modules integrate smoothly.

2.6 Extreme Programming (XP) Methodology

2.6.1 Extreme Programming Methodology – Design

An agile framework focused heavily on software engineering best practices like pair programming, test-driven development (TDD), and continuous integration.

2.6.2 Advantages of Extreme Programming Model

Produces exceptionally high-quality code; highly responsive to evolving customer demands.

2.6.3 Disadvantages of Extreme Programming Model

High resource and cost overhead; demanding on developers, which can lead to rapid burnout.

2.7 Our Project Methodology

For the Grambandhan platform, a structured yet iterative approach based on the standard Software Development Life Cycle (SDLC) will be utilized. Because Grambandhan involves complex integrations like digital payments, AI risk analysis, and multi-actor interfaces (Farmers, Investors, Admins, Field Agents), an iterative SDLC ensures early requirements are met while allowing for refinement.

2.8 Methodology Phases

- **Requirement Gathering:** Collected data through surveys, research, and problem analysis in rural sectors.
- **System Design:** Created system architecture, database design, and UI mockups utilizing Figma.
- **Analysis:** Compared existing platforms and identified missing features in the market.
- **Development:** Built frontend utilizing HTML/CSS/JavaScript and backend utilizing PHP/Node.js with MySQL.
- **Testing:** Performed functional, usability, and security testing across modules.
- **Deployment:** Deployed on a web server for reliable user access.
- **Maintenance:** Continuous updates, bug fixing, and feature improvements to support growing users.

3. Work Breakdown Structure (WBS)

3.1 What is Work Breakdown Structure (WBS)

A Work Breakdown Structure (WBS) is a hierarchical, deliverable-oriented decomposition of a project into smaller, more manageable components. It visually defines the total scope of the project, mapping out every task required to achieve the final objectives.

3.2 WBS Design Principles

- **The 100% Rule:** The WBS must encompass exactly 100% of the project scope, including internal, external, and interim deliverables.
- **Planned Outcomes, Not Planned Actions:** Elements should be defined by their resulting deliverables (nouns) rather than the actions needed to create them (verbs).
- **Level 2 is the Most Important:** The second level dictates the strategic framework of the project and must logically encompass all lower-level items.
- **The Four Elements in Each WBS Element:** Every element should be defined by its scope, schedule, budget, and assigned responsibility.
- **Mutually Exclusive Elements:** There must be no overlap between elements to avoid duplicated effort and ambiguous accountability.
- **The 40-Hour Rule of Decomposition:** Work packages should typically take no less than 40 hours to complete to avoid micromanagement.
- **The 4% Rule of Decomposition:** No single work package should account for less than 4% of the total project effort.

3.3 WBS of Our Project

- **1.0 Grambandhan Platform**
  - **1.1 Planning & Analysis**
    - **1.1.1** Target audience surveys (Farmers, Investors)
    - **1.1.2** Existing platform benchmark analysis
    - **1.1.3** Requirements specification drafting
  - **1.2 System Design**
    - **1.2.1** UI/UX prototyping via Figma
    - **1.2.2** Database architecture (MySQL)
    - **1.2.3** UML Diagram creation (DFD, Class, Swimlane)
  - **1.3 Module Development**
    - **1.3.1** User Registration & Role Management
    - **1.3.2** Project Listing & Investment System
    - **1.3.3** Real-Time Progress Tracking Dashboard
    - **1.3.4** Online Marketplace Integration
    - **1.3.5** AI-Based Risk Suggestion Model
    - **1.3.6** Digital Payment (bKash, Nagad) Integration
  - **1.4 Testing & Deployment**
    - **1.4.1** Security and fraud protection testing
    - **1.4.2** Cross-platform web and mobile compatibility testing
    - **1.4.3** Web server deployment and database hosting

4. Wideband Delphi

4.1 What is Wideband Delphi

Wideband Delphi is an iterative, consensus-based estimation technique where a panel of experts anonymously estimates project effort or cost. It minimizes individual bias and leverages collective expertise to produce highly accurate resource and timeline forecasts.

4.2 Wideband Delphi Process

- **Choose the Team:** Select a moderator and a diverse group of 3-7 experts (developers, designers, testers) familiar with the project scope.
- **Kick-off Meeting:** The moderator presents the project requirements, ensuring all estimators share a unified understanding of the deliverables.
- **Individual Preparation:** Estimators independently generate their effort or cost estimates without discussing them with others.
- **Estimation Session:** Anonymous estimates are collected, plotted on a chart, and discussed—especially the outliers—to clarify misunderstandings.

- **Assemble Tasks:** The team breaks down complex features into detailed tasks during discussions to refine their understanding.
- **Review Results:** The cycle is repeated until the team reaches a reasonable consensus on the final estimate.

4.3 Assemble Table

Below is a synthesized estimation table for the core modules of Grambandhan.

Task ID	Task Description	Est. Hours	Assigned To
WD-01	Develop User Registration & Role-Based Verification	45	Saptarshi
WD-02	Build Project Listing & Profit-Sharing System	60	Sourav
WD-03	Implement Real-Time Tracking (Images, Video, Reports)	55	Mahin
WD-04	Integrate Online Marketplace & Cart	65	Partha
WD-05	Develop AI-Based Risk Suggestion Algorithm	70	Jony
WD-06	Integrate Payment Gateways (bKash, Nagad) & Escrow	50	Saptarshi
WD-07	Implement Fraud Protection & Verification Module	40	Sourav

Table 1: Wideband Delphi Task Estimation for Grambandhan

5. Risk Management

5.1 What is Risk Management

Risk Management is the continuous process of identifying, analyzing, evaluating, and responding to potential threats that could negatively impact a project's objectives, timeline, or budget.

5.2 Risk Mitigation, Monitoring, and Management (RMMM)

5.2.1 Risk Mitigation

Proactive steps taken before a risk occurs to reduce its likelihood or financial impact.

5.2.2 Risk Monitoring

Continuously tracking identified risks, monitoring residual risks, and identifying new threats throughout the project lifecycle.

5.2.3 Risk Management and Planning

Establishing formal contingency plans and defining strict response protocols to be executed if a risk materializes.

5.3 Risk Plan

The Grambandhan platform faces unique operational and technical risks given its target demographic and financial nature.

- **Digital Literacy Limitations:** Rural users may struggle with limited technical knowledge, necessitating a simple and easy-to-use interface along with training awareness.
- **Connectivity Dependency:** The system's dependency on internet connectivity requires designing lightweight applications that function reliably on minimum 2 Mbps networks.
- **Trust and Security Issues:** Initial trust issues and managing fraud are critical risks; these are mitigated by implementing rigorous user verification and the Fraud Protection System.
- **Payment Provider Dependency:** Integration depends heavily on third-party mobile payment services like bKash and Nagad. System architecture will incorporate multiple financial gateways to prevent single points of failure.
- **Agricultural Risks:** Since farming projects carry inherent uncertainty due to weather and pests, the platform integrates an AI-based Risk Suggestion feature and an Insurance & Risk Coverage system to protect stakeholders from catastrophic losses.